

Individual Contribution Report — Lab 9 Phase 5

Team Lead / Data & Back-End Engineer: Daniel Evans

1. Contributions Overview

I focused on the Phase 4 unified dashboard integration, Phase 2 pipeline benchmarking, and collaborative documentation for the Research-A-Thon submission.

2. Phase 4 Dashboard (Repo 1) — 100% Complete

I developed the unified Streamlit dashboard for the `CS5542_SmartSC_Optimization_System` repository, merging analytics features from Lab 6 with the new CPP safety protocol from Lab 8.

- **Persistent Sidebar:** Implemented a global navigation sidebar with project description, live data-freshness timestamps, and a ‘Reset Session’ button for better UX.
- **Reasoning Path Expander:** Created a collapsible UI component that displays the underlying ReMindRAG traversal sources and the CPP Compliance Agent’s decision (PASS/VETO).
- **Operational Analytics:** Successfully ported and integrated the Fleet, Routes, Fuel, and Safety analytics tabs from Lab 6 into the unified dashboard.
- **Stability & Resilience:** Applied exponential-backoff retry decorators to all Snowflake Cortex and external API calls to prevent dashboard failures during rate-limiting events.

3. Phase 2 Benchmarking (Repo 3) — 100% Complete

I executed the pipeline evaluation runs for the `ReMindRAG_Week7` reproducibility project and committed the results.

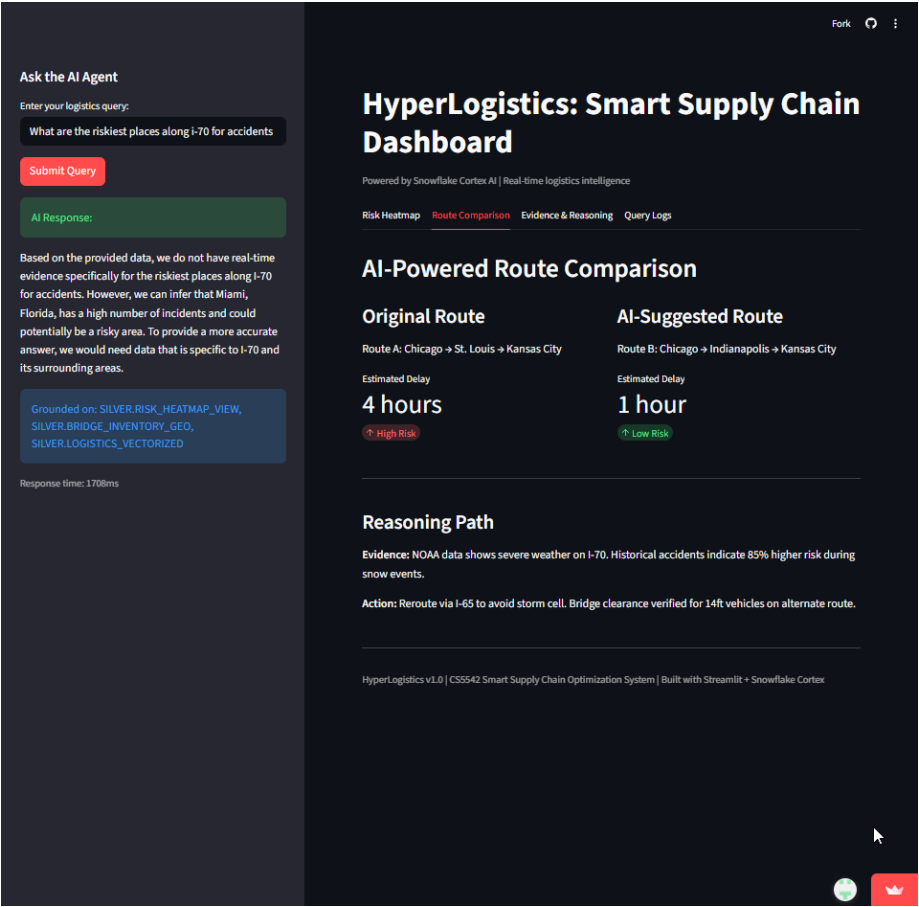
- **LooGLE Evaluation:** Ran 5-title subset runs with an Avg F1 of 0.7125.
- **HotpotQA Evaluation:** Ran 50-question subset tests with an Avg F1/Accuracy of 0.7600.
- **Results Logging:** Recorded all scores in the project root `eval_results.json` for CI/CD integration.

4. Commits & Evidence

Repository	Commits / Artifacts
CS5542_SmartSC_Optimization_System	src/app/dashboard.py, CONTRIBUTIONS.md, README.md
ReMindRAG_Week7	eval_results.json, tests/test_repro_variance.py
COMP_SCI_5542	Week_9/reports/CONTRIBUTION_DE.md

Visual Evidence (Phase 2 Integration)

Unified dashboard showing the ReMindRAG reasoning path and evidence injection:



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Verified by: Daniel Evans Date: 2026-03-24